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$$f(x) = 3x - x \ln x^3$$

$$\begin{aligned} f'(x) &= \frac{d}{dx}(3x) - \frac{d}{dx}(x \ln x^3) \\ &= 3 - \left(\frac{d}{dx}(x) \cdot \ln x^3 + x \cdot \frac{d}{dx}(\ln x^3) \right) \\ &= 3 - \left(\ln x^3 + x \cdot \frac{1}{x^3} \cdot \frac{d}{dx}(x^3) \right) \\ &= 3 - \ln x^3 - \frac{3x^2}{x^2} \\ &= -\ln x^3 \end{aligned}$$

References